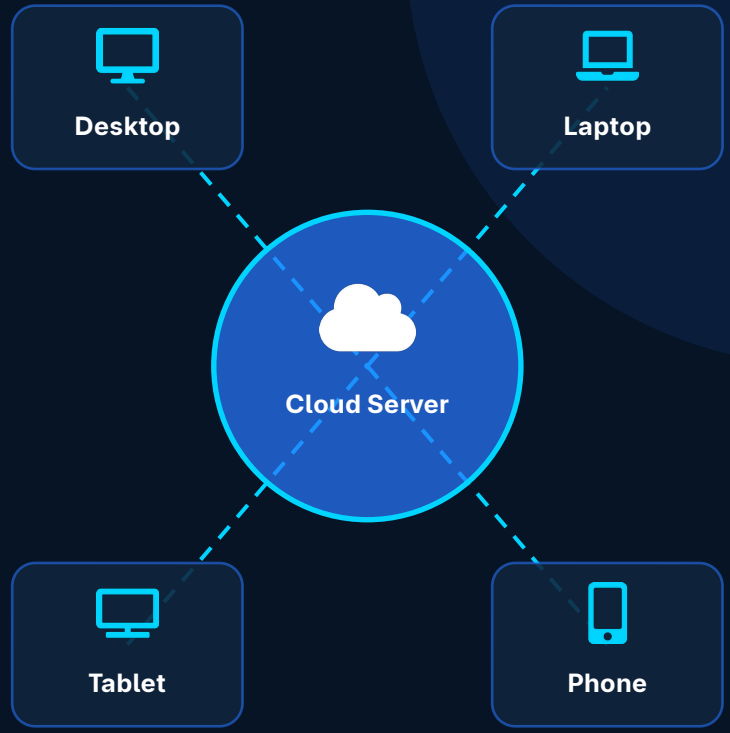
CLOUD COMPUTING

Desktop Virtualization

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Subject: Cloud Computing

College: JD College of Engineering & Management

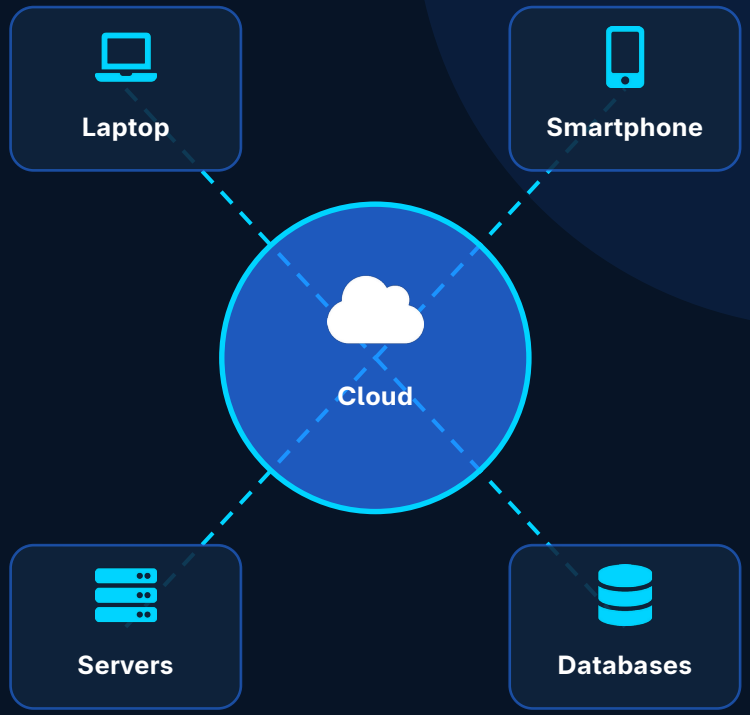


Introduction to Cloud Computing

Cloud computing: delivery of computing resources over the internet

Resources: servers, storage, networking, databases and software

Benefits: scalability, flexibility and cost efficiency



Popular providers

AWS | Microsoft Azure | Google Cloud

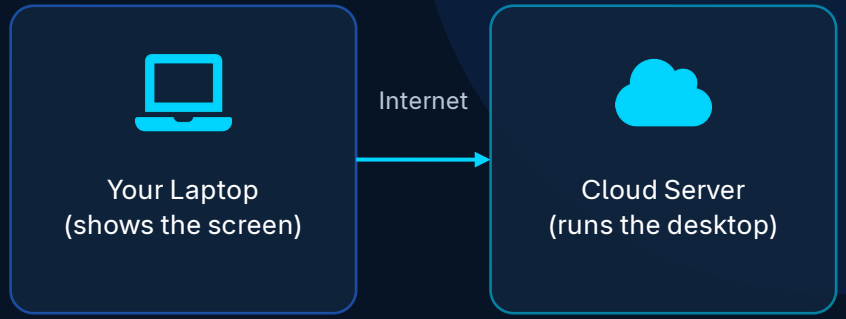
What is Desktop Virtualization?

Separates the desktop environment from the physical device

Hosts desktops on central servers or cloud

Remote access from almost any device

Centralized management of apps and user data



Real-life example
Open your college lab desktop from home on your own laptop. Same files, same apps.

Virtual Desktop
OS, apps and data live on the server. The device only displays the output.

How Desktop Virtualization Works



INPUT: keyboard and mouse commands

OUTPUT: screen image sent back

1. Desktop created on server
2. OS and apps run virtually
3. User connects from a device
4. Inputs sent to the server
5. Screen output sent back

Types of Desktop Virtualization



VDI

Virtual Desktop Infrastructure

Each user gets a dedicated VM on company servers.



DaaS

Desktop as a Service

A cloud provider hosts and manages the desktops.



RDS

Remote Desktop Services

Many users share one server OS through sessions.



Local

Local Desktop Virtualization

A hypervisor on your own PC runs guest desktops.

Virtual Desktop Infrastructure (VDI)

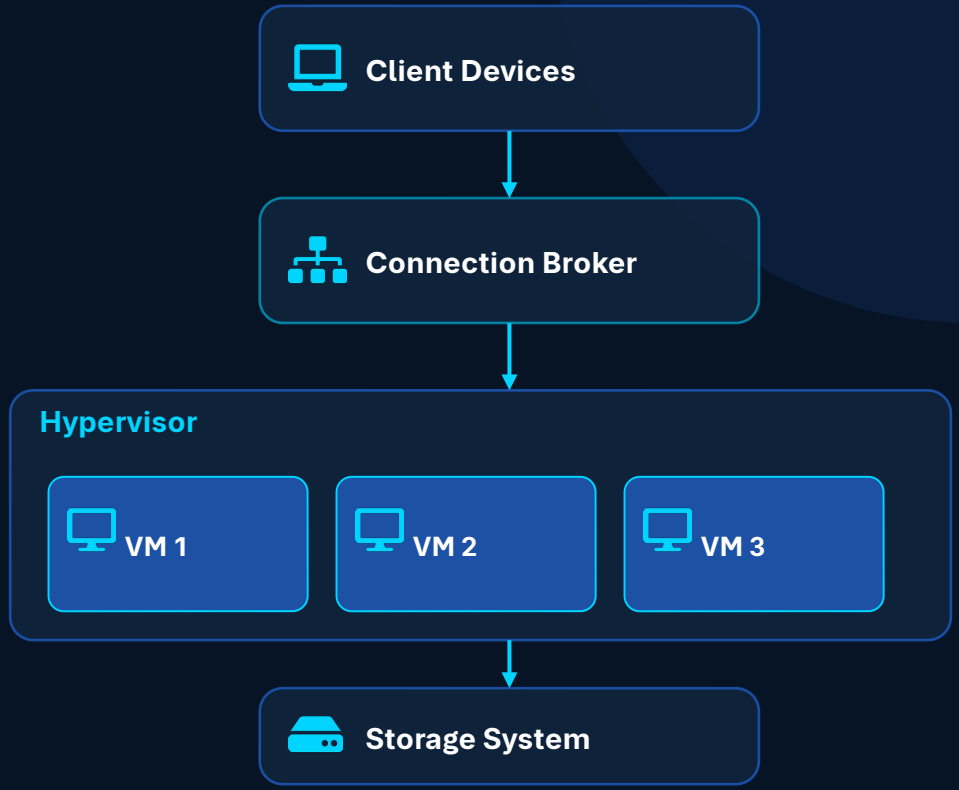
Centralized virtual desktop hosting

One VM per user

Hypervisor manages the VMs

Admins control everything centrally

Remote desktop access



Desktop as a Service (DaaS)

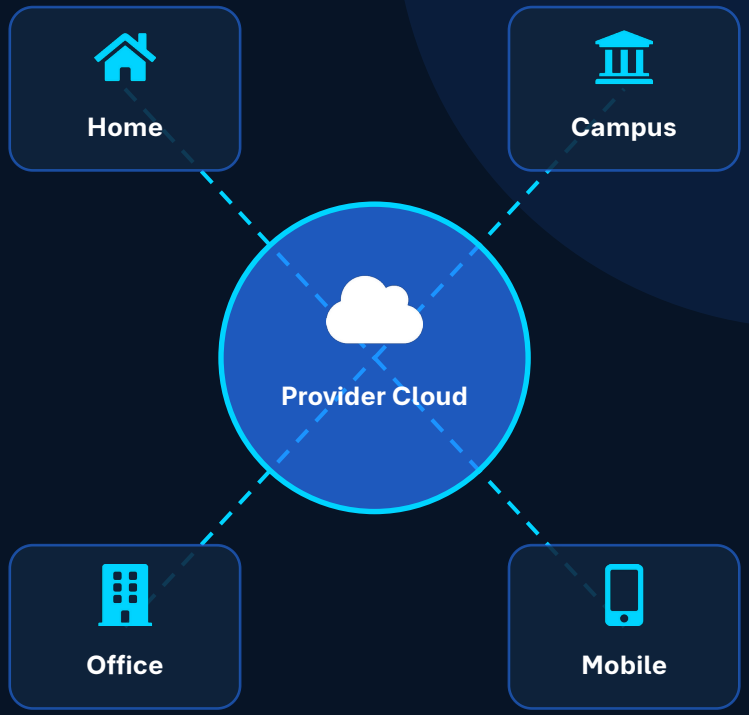
Cloud-hosted virtual desktops

Provider manages the infrastructure

Pricing by subscription or usage

Access over the internet

Less infrastructure maintenance



Examples

Amazon WorkSpaces | Azure Virtual Desktop | Citrix DaaS

Key Components of Desktop Virtualization



1. Hypervisor

Creates and manages virtual machines.



2. Virtual Machines

Provide independent desktop environments.



3. Connection Broker

Connects users to their assigned desktops.



4. Storage

Stores operating systems, applications and user data.



5. Network

Enables communication between client and server.



6. Client Device

Gives the user access to the virtual desktop.

Benefits of Desktop Virtualization



Cost Reduction

Less hardware spending and lower upkeep.



Centralized Management

Update and manage desktops from one place.



Remote Accessibility

Work from anywhere with internet.



Scalability

Add or remove desktops as needed.



Data Security

Data stays on central servers, not devices.



Flexibility

Use many devices with the same desktop.

Challenges and Limitations



Internet Dependency

A stable connection is needed to work.



Initial Setup Cost

Servers and licenses need upfront investment.



Network Latency

Slow networks make the desktop laggy.



Security Risks

Central servers become a target to protect.



Resource Management

Shared CPU, RAM and storage must be balanced.



Technical Complexity

Needs skilled staff to design and maintain.

Desktop Virtualization vs Traditional Desktop



Traditional: OS, apps and data on one local PC



Virtual: desktop runs on a server, any device shows it

Traditional Desktop	Virtual Desktop
Runs on physical hardware	**Runs on virtual infrastructure**
Local data storage	**Centralized storage possible**
Limited remote access	**Remote access supported**
Individual maintenance	**Centralized management**
Hardware upgrades needed	**Resources can be scaled**
Device-dependent	**Device-independent access**

Applications of Desktop Virtualization



Education

Virtual computer labs for students.



Business

Remote working environments.



Healthcare

Access to applications across departments.



IT Industry

Software development and testing.



Banking

Centralized desktop management.



Call Centers

Centrally managed employee desktops.

Popular Desktop Virtualization Platforms

Omnissa Horizon

PROVIDER

Omnissa

Virtual desktop and app delivery platform.

Amazon WorkSpaces

PROVIDER

Amazon Web Services

Managed cloud desktops from AWS.

Azure Virtual Desktop

PROVIDER

Microsoft

Cloud-hosted Windows desktops and apps on Azure.

Citrix DaaS

PROVIDER

Citrix

Cloud-delivered virtual apps and desktops.

Future Scope of Desktop Virtualization

AI Optimization

Smarter allocation of resources.

Edge Computing

Desktops closer to users, lower delay.

5G Connectivity

Faster, more reliable remote access.

Zero Trust Security

Verify every user and device.

Hybrid Cloud

Mix on-premises and cloud desktops.

Automation

Automatic setup, scaling and updates.



Conclusion & References

Important technology in cloud computing

Centralized management and remote access

Less hardware dependency

Supports businesses and education

Evolving with AI, 5G and cloud

References

- Microsoft Learn: Azure Virtual Desktop
- AWS Documentation: Amazon WorkSpaces
- Citrix Documentation
- Omnissa Documentation
- NIST Cloud Computing Resources

THANK YOU!

Questions & Answers